

Mika Be

Making difficult systems usable.

I build the software around technical capability: research tooling that preserves equation structure, local-first products that survive real use, and simulations people can inspect instead of simply trust.

APPLIED AI

SOFTWARE ENGINEERING

PRODUCT-MINDED SYSTEMS

01

Equation-aware speech

Python tooling + IEEE ISEC 2026

02

Local-first editing

Flutter + Drift + Firebase

03

Inspectable simulation

JavaScript + MNA + tests

Evidence, with the status left intact.

The following pages separate contribution from product status. Real captures are used where available. 'Private beta' and 'active development' are deliberate boundaries, not missing polish language.

CURRENT TARGET

Summer 2027
applied AI and
software
engineering
internships

Scientific speech needs structure, not just a voice.

2ND PLACE BEST FULL PAPER, IEEE ISEC 2026

THE PROBLEM

Legacy scientific PDFs often flatten equations into unreadable fragments. The useful problem is preserving enough structure that a listener can follow the math.

MY CONTRIBUTION

- Co-authored the IEEE ISEC 2026 paper and contributed to the accessibility-focused research workflow.
- Built Python tooling to extract tagged narration from LaTeX and apply structural derivative, ratio, and pacing rewrites.
- Connected prepared narration to Piper synthesis and produced WAV outputs with traceable CSV manifests.

WHAT THE EVIDENCE SUPPORTS

A reproducible local speech-preparation and synthesis workflow. This packet does not claim a deployed service, custom model training from the current workspace, or production-scale inference.

Kuiper TTS:
Making Legacy Scientific Papers Accessible by
Converting Equations into Spoken Language

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Abstract - Our project aims to create a tool that is easy for any person to use to enjoy the beauty of mathematics and science. Scientific papers are often inaccessible to low vision readers because legacy PDFs expose little semantic structure for screen readers, and the problem is amplified by mathematical and scientific notation. For many other readers, challenges such as dyslexia, difficulty in learning symbols, or having a limited mathematical background also create a barrier to learning. Meanwhile, standard PDF-to-text or Optical Character Recognition (OCR) tools available often omit equations or reduce them to unreadable fragments. Current screen-readers are extremely difficult to use, requiring expertise, and are unreliable at reading equations. We present a user-focused system that shifts the focus from screen reader to an audio player capable of reading an ordinary text file and reading math and scientific content aloud with high fidelity. In this paper, we describe a method to prepare a LaTeX document in a batch from PDF in advance and other sources, mindfully reducing power consumption. The audio player will be capable of functioning for hours on a small platform such as a cell phone, will not require an internet connection, and will be freely available and open source.

Index Terms - Mathematics Education, Text-to-Speech, Accessibility, Math-to-Speech, LaTeX, MathML, LLM, PDF, OCR, Optical Character Recognition.

INTRODUCTION
The body of human knowledge – particularly in technical fields such as mathematics and engineering – resides largely within curated repositories of formal literature: digital archives, scholarly libraries, and structured electronic publications. Particularly in technical topics, most documents are formatted in Portable Document Format (PDF) and legacy formats. PDF provides a single visual format portable across platforms, but is not well-designed for the purpose of disseminating equations and other artifacts, nor can it be easily read by a screen reader. There are also documents remaining in print and handwriting that remain unscanned. Newer scientific papers are often in either LaTeX or Word, preserving more of the semantic content. Our Kuiper-TTS project includes a software

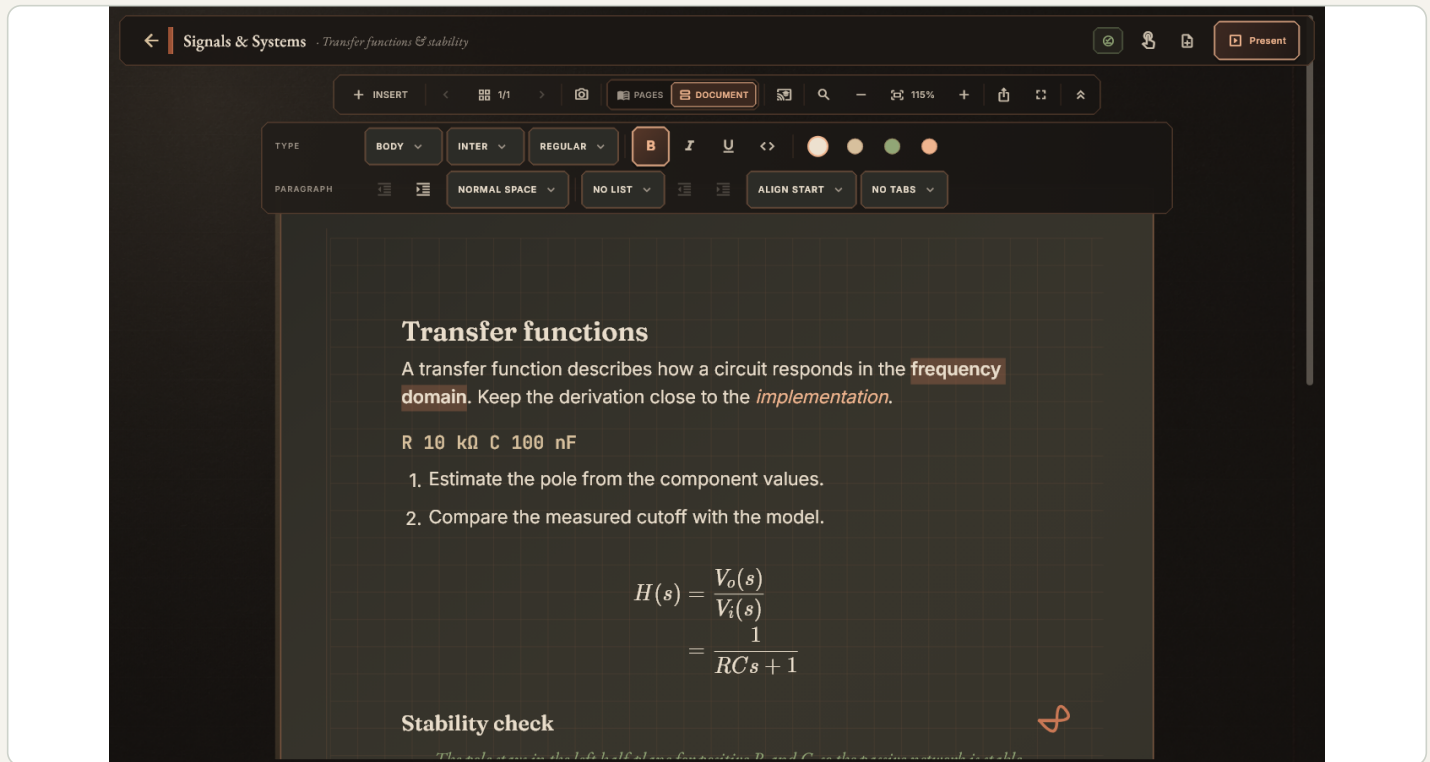
component to read a text file as audio, and a set of agents and AI processors that convert legacy scientific papers into an intermediate format where the equations, figures, and images have been validated, and spoken equivalents pre-computed. The result is a document that can be played in audio very efficiently. Because the presentation of equations in PDF can be ambiguous, preprocessing with multiple agents and AI filters is used to try to preserve the semantic content and store it in a precise form. We use LaTeX as the intermediate form, embedding any content for the audio player in tags that identify how it is to be handled. The result is an audio player that can read a normal text file, but can also read a scientific paper with equations. The benefit is to provide for low-vision users, a hands-free application (such as driving) or an augmented reading experience for people with dyslexia, and for validating input to ensure AI imports the content correctly for retrieval-augmented generation (RAG). Our experiments show that many AIs trained to perform OCR on PDF files will fail to read equations at the top and bottom of the page because they must reject erroneous header and, for example, footer text. Because of this, it seems likely that many AIs are being trained on incomplete information. The pipeline we have defined to read, edit, and correct PDF files and save the results may potentially increase the accuracy of LLMs trained from scientific papers, though this has not yet been tested. Scientific knowledge is overwhelmingly distributed as PDFs, including decades of archival proceedings, technical reports, and scanned journals [1-5]. While PDFs preserve visual layout, they expose weak semantics: screen readers often follow a flawed linear order, drop structure, and render mathematics as fragmented or unusable for text-to-speech. This also means that AI OCR correction is imperfect.

EVIDENCE

Paper · source code · saved speech output
Award listing · experiment manifests

One page model. Several ways to think.

PRIVATE BETA



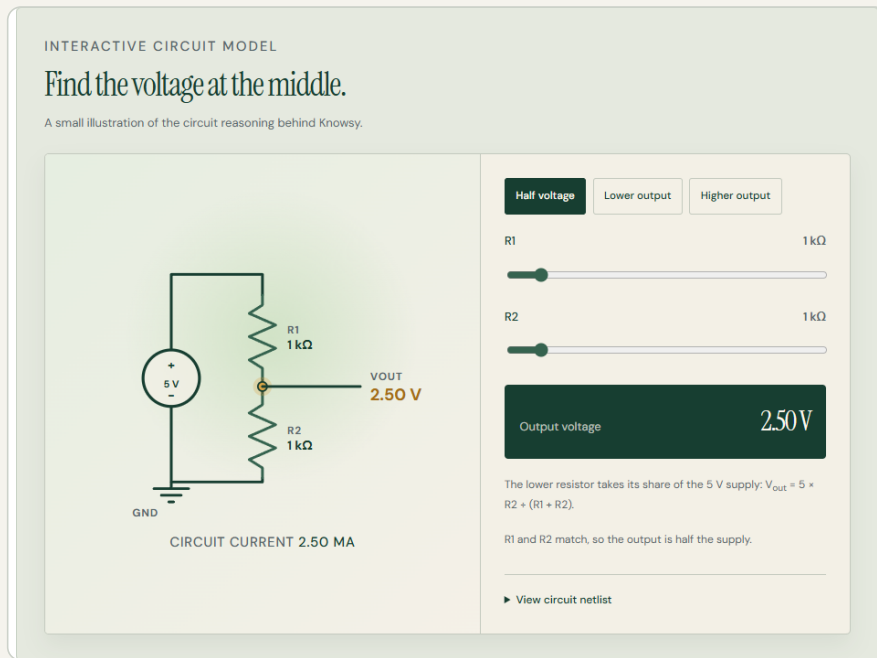
WHAT I BUILT

- A Flutter editor where text, ink, equations, code, images, and diagrams share a stable 900 x 1200 coordinate space.
- Device-specific interfaces over the same document model, so desktop editing, iPad writing, and phone capture stay coherent.

SYSTEMS JUDGMENT

- Local-first persistence with Drift/SQLite and durable Firebase synchronization for notebooks, media, and revisions.
- Explicit beta status: current source and captures support the architecture and interface; sustained Apple Pencil feel remains a physical-device gate.

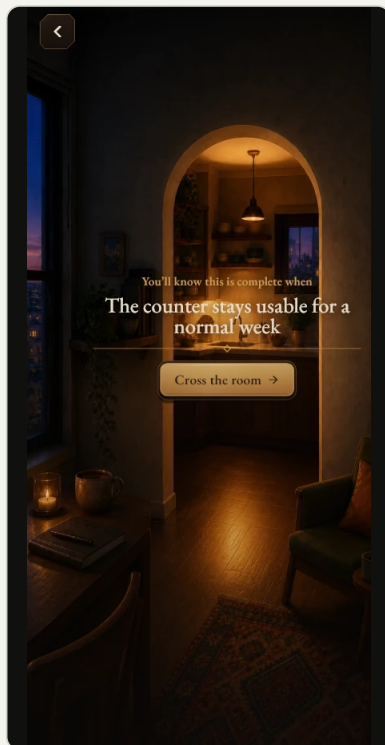
Make the model inspectable.



KNOWSY

Circuit reasoning you can touch.

- Led a 3-person breadboard-module effort.
- Built wiring validation, digital simulation, and inspectable netlists.
- Implemented modified nodal analysis with complex admittances and frequency sweeps.



ROOM OF DAYS / ACTIVE DEVELOPMENT

Reliable beneath the atmosphere.

A local-first Flutter product for quests, goals, planning, and reflection. I own the product direction, interaction design, domain model, persistence, and cross-platform implementation.

- Offline core with validated JSON persistence and backups.
- Optional Firebase sync rather than a network-dependent product shell.
- Targets Android, iOS, and web; no unverified active-user count is used here.